

Heightfield Construction

$$h = z_s \tilde{g}_t$$

Registration

$$\mathbf{W} = \mathbf{U}\Sigma\mathbf{V}^\top$$

Matching

$$|\mathbf{u}^\top \mathbf{F} \mathbf{u}| < \varepsilon$$

Fusion

$$\mathbf{X} = \min \|\mathbf{A}\mathbf{X}\|$$

(a) Input frame

(b) Gradient field

(c) Heightfield mesh

(d) Registration (epipolar lines)

(e) Dense matching (L-R)

(f) Fused point cloud

